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 <https://github.com/Leilazehui/Leilazehui.github.io>
Space Systems, Aerospace Structures, UAV Design, CFD, MDO

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RESEARCH EXPERIENCES & MANUSCRIPTS

Multidisciplinary Design Optimization of a Fixed-Wing Aircraft or AIAA DBF Mission Scoring

Hong Kong

First Author (Expected Submission: July 2026)

Nov 2025 - now

- A parametric Multidisciplinary Optimization (MDO) framework was developed in MATLAB using a deterministic grid-search approach to explore the design space systematically to obtain the predicted performance of the preliminary design that maximizes mission score while satisfying aerodynamic and performance constraints.

MAJOR AEROSPACE PROJECTS & COMPETITIONS

HKU Design, Build and Fly Team

Hong Kong, USA

HKU AIAA DBF Competition Team Leader

Aug 2025 – now

- Optimized main wing specification under the constraint of 1.25 and 1.5-meter wingspan, to provide sufficient lift for an 8kg and 2.7kg aircraft and fast-speed flight.
- Conducted Aerodynamic analysis on airfoils and wings on XFLr5 to verify the stability of the aircraft and ease the manufacturing process.
- Performed CFD analysis on Ansys on aircraft model and checked the pressure distribution on the aircraft to improve aircraft aerodynamics.
- Designed banner mechanisms by utilizing CAD, Arduino and 3D printing for prototypes, deployed and released 4-m long banner in flight.
- Designed, examined and tested a Multidisciplinary Optimization Framework (MDO) by automating 100 iterations for mission score optimization and selected the best score case for the team aircraft configuration on MATLAB.
- Ranked 32nd out of 98 teams from universities worldwide competing in AIAA DBF 2026, 3rd in Asia and 2nd in Hong Kong

HKU DBF Student Interest Group Leader

Sept 2024 – now

- Conducted over 30 workshops and training to promote the team to new members, and received 100K HKD funding from Innovation Academy, HKU.
- Secured three industry sponsorships, including Ansys, L’Voyage and the Safety Collaborative for DBF2025-2026.
- Manage the team’s human resources, documents, promotion and train next year’s leaders to enlarge the team size and development.
- Conducted research study on 3D printing material regarding the topic of lightweight design and analysis for ribs construction in fixed-wing model aircraft, tested around 5-8 materials and provide the most ideal solution for lightweight fixed-wing design.
- Designed team jacket and posters for DBF 2025 & 2026

MATLB Interstellar Project

Dec 2024

- Analyze stellar motion by data plotting regarding intensity and wavelength of 7 stars to calculate their doppler effect, and compare their stellar spectra provided in the database.

RAICOM Robotic Arm Challenge (Shenzhen)

Jul 2025

- Broke down the structure of the algorithm for automated and semi-automated missions; performed data annotation for AI model training.

EDUCATION

The University of Hong Kong, Bachelor’s in Mechanical Engineering

2022 – now

- Highest Attained GPA: 3.36/4.3
- Cumulative GPA: 2.76/4.3
- Scholarship: Tam Wing Fan Innovation Academy Funding 2025 – 2026: 100K HKD
- Course taken: Mechanics of Solid (A), Engineering design and manufacture (A+), Unmanned Aerial Vehicle design (A+), Dynamics and Control (A-), Thermodynamics (A-)

WORK & RESEARCH EXPERIENCES

The Hong Kong Productivity Council

Hong Kong

Winter Intern in Smart Manufacturing Division

Dec 2025 – Jan 2026

- Researched global textile recycling rules and developed an ESG toolkit to drive circular sustainability for Hong Kong fashion firms, supporting local SMEs in green recycling and AI integration. Also surveyed AI applications in footwear operations.
- Designed a 3D camera plus multi-axis robotic arm mounting system, applying 2D, 3D drawing, BOM and GD&T standard for manufacturing via SolidWorks.
- Trained an AI model on Roboflow to classify normal and abnormal temperature regions on a robotic arm
- Designed a conceptual drone inspection project using SLAM, visual inspection, and FEA data to detect internal building cracks.

Faculty of Engineering, the University of Hong Kong

Hong Kong

Part-time Temporary Student Research Assistant

Sept 2025 – Apr 2026

- Mentored over 30 students in robotic arm control and design, enhancing their practical skills and understanding of engineering concepts.
- Provide constructive feedback to lecturers on behalf of workshop content and delivery which assisted them in workshop development.

Feelings Group Limited | Hong Kong Science & Technology Park

Student Technician – Hardware team

Hong Kong

Dec 2024 – Jan 2025

- Provide academic and industrial research, market database radar hardware for medical devices tracking vital signs of small animals.
- Assisted in camera development using electronic modules, such as Raspberry Pi, LAN network and camera module for CCTV development for the medical device.

The Hong Kong and China Gas Company Limited (Towngas)

Summer intern in residential product & development department

Hong Kong

Jul - Aug 2024

- Contributed to the development of 7 IoT gas appliances, and 10+ gateways.
- Identified potential problems in functional tests and provided effective suggestions to enhance user-friendliness of the mobile app interface.
- Assisted in the assessment of gas appliance accessories such as flexible hose and gas malleable iron fittings.
- Analyzed market potential for company's products and participated in daily meetings, contributed to strategic planning and understanding team structure and working principles and provided valuable feedback to supervisor regarding the market demand and daily workflow

CERTIFICATES & AFFILIATIONS

Royal Aeronautical Society (HK) Student Member 2025

IMEchE Affiliated Member & Committee 2025

General Secretary, Publication and Publicity Secretary, Physics Society, HKU (2023)

HKIE Aircraft Division – Student Member 2025 - 2026

Peking University Summer Exchange Program 2023

Junior Student Ambassador of Equal Opportunity Unit, HKU 2023

HKU Lead-For-Life Cohort Member 2022 – 2026

Champion in Public Group, The 1st “National Aerospace Dream”, Hong Kong Aerospace Knowledge Competition 2022

SKILLS

Language: Chinese (Mandarin, Cantonese): Native, English: Proficient, German: Basic

CAD Software: SolidWorks, AutoCAD, fusion360

CFD Software: Xflr5, Ansys

Other Software: HIKIMICRO Studio, Bambu lab Studio, Microsoft Office Suite, Roboflow

Programming: MATLAB, Arduino, Python, Orange data mining, Visual studio